YELABUGA TATARSTAN, Russian Federation

WMO: **285060**

Lat: **55.759N** Lon: **52.044E** Elev: **298** StdP: **14.54** Time Zone: **3.00 (E03)** Period: **94-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR 99.6%			99%			Coldest Month WS/MCDB 0.4%		1%		MCWS/PCWD to 99.6% DB		WSF	
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	**1**	**-19.2**	**-13.2**	**-24.8**	**1.4**	**-18.7**	**-18.6**	**2.0**	**-12.8**	**29.3**	**19.6**	**23.6**	**20.8**	**4.8**	**230**	**0.567**	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB 0.4%		1%		2%		Evaporation WB/MCDB 0.4%		1%		2%		MCWS/PCWD to 0.4% DB		
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	**7**	**17.6**	**88.4**	**68.7**	**84.6**	**68.0**	**81.2**	**65.9**	**71.1**	**83.8**	**69.3**	**81.4**	**67.7**	**78.6**	**5.4**	**210**	(2)

	Dehumidification DP/MCDB and HR 0.4%			1%			2%			Enthalpy/MCDB 0.4%		1%		2%		Extreme Max WB	
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	**67.0**	**100.3**	**76.2**	**65.2**	**94.1**	**74.6**	**63.5**	**88.5**	**73.5**	**35.1**	**84.1**	**33.6**	**82.0**	**32.2**	**78.8**	**87.8**	(3)

Extreme Annual Design Conditions

	Extreme Annual WS 1%	2.5%	5%		Extreme Annual Temperature Mean Min	Mean Max	Standard Deviation Min	Standard Deviation Max	n=5 years Min	n=5 years Max	n=10 years Min	n=10 years Max	n=20 years Min	n=20 years Max	n=50 years Min	n=50 years Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	**23.2**	**19.7**	**17.1**	DB	**-24.4**	**91.6**	**7.9**	**4.2**	**-30.1**	**94.6**	**-34.7**	**97.1**	**-39.1**	**99.4**	**-44.8**	**102.5**	(4)
(5)				WB	**-24.7**	**74.6**	**7.8**	**4.7**	**-30.3**	**77.9**	**-34.8**	**80.7**	**-39.2**	**83.3**	**-44.9**	**86.7**	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	**Temperatures, Degree-Days and Degree-Hours**		DBAvg	**40.1**	**11.5**	**12.5**	**24.8**	**41.4**	**55.6**	**64.1**	**68.2**	**63.9**	**53.6**	**40.7**	**26.9**	**16.1**	(6)
(7)			DBStd	**22.53**	**12.90**	**12.88**	**8.76**	**10.07**	**9.27**	**7.95**	**7.02**	**7.88**	**8.24**	**8.39**	**10.65**	**12.98**	(7)
(8)			HDD50	**5484**	**1193**	**1049**	**783**	**295**	**53**	**5**	**0**	**1**	**52**	**308**	**695**	**1050**	(8)
(9)			HDD65	**9442**	**1658**	**1469**	**1248**	**709**	**313**	**111**	**49**	**119**	**354**	**753**	**1144**	**1515**	(9)
(10)			CDD50	**1864**	**0**	**0**	**0**	**36**	**227**	**427**	**565**	**430**	**160**	**19**	**0**	**0**	(10)
(11)			CDD65	**349**	**0**	**0**	**0**	**0**	**21**	**84**	**149**	**83**	**12**	**0**	**0**	**0**	(11)
(12)			CDH74	**3258**	**0**	**0**	**0**	**6**	**244**	**747**	**1398**	**774**	**89**	**0**	**0**	**0**	(12)
(13)			CDH80	**1014**	**0**	**0**	**0**	**0**	**35**	**217**	**468**	**279**	**15**	**0**	**0**	**0**	(13)
(14)	**Wind**		WSAvg	**6.3**	**6.4**	**6.3**	**6.7**	**6.8**	**6.8**	**5.9**	**5.0**	**5.5**	**5.7**	**7.0**	**6.8**	**6.3**	(14)
(15)	**Precipitation**		PrecAvg	**21.4**	**1.5**	**1.1**	**1.1**	**1.6**	**1.8**	**2.0**	**2.8**	**2.2**	**2.3**	**2.2**	**1.7**	**1.5**	(15)
(16)			PrecMax	**30.4**	**3.1**	**3.0**	**3.5**	**5.5**	**4.3**	**4.3**	**6.7**	**5.9**	**6.3**	**4.4**	**4.1**	**3.0**	(16)
(17)			PrecMin	**14.3**	**0.2**	**0.0**	**0.0**	**0.4**	**0.2**	**0.2**	**0.9**	**0.3**	**0.2**	**0.0**	**0.4**	**0.1**	(17)
(18)			PrecStd	**4.5**	**0.8**	**0.7**	**0.7**	**1.1**	**1.0**	**1.0**	**1.5**	**1.3**	**1.7**	**1.1**	**0.9**	**0.8**	(18)
(19)	**Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures**	0.4%	DB		**36.0**	**36.4**	**48.2**	**74.5**	**83.7**	**90.7**	**93.3**	**94.2**	**82.1**	**66.7**	**51.3**	**37.7**	(19)
(20)			MCWB		**34.2**	**33.6**	**41.3**	**55.8**	**63.7**	**68.3**	**69.2**	**70.7**	**62.0**	**54.1**	**49.0**	**35.6**	(20)
(21)		2%	DB		**33.7**	**34.8**	**41.8**	**68.1**	**80.3**	**85.7**	**88.7**	**87.2**	**76.0**	**61.5**	**46.1**	**35.6**	(21)
(22)			MCWB		**32.4**	**32.8**	**36.6**	**52.6**	**61.8**	**67.9**	**68.9**	**68.5**	**59.4**	**51.6**	**43.4**	**34.0**	(22)
(23)		5%	DB		**31.5**	**33.1**	**38.4**	**63.2**	**76.5**	**81.8**	**85.2**	**82.0**	**71.4**	**56.7**	**41.9**	**33.9**	(23)
(24)			MCWB		**30.3**	**31.6**	**34.6**	**50.2**	**59.9**	**66.0**	**69.0**	**66.7**	**58.6**	**49.6**	**40.0**	**32.5**	(24)
(25)		10%	DB		**28.8**	**30.2**	**36.0**	**57.7**	**72.1**	**78.1**	**81.6**	**77.4**	**66.5**	**52.6**	**38.5**	**32.1**	(25)
(26)			MCWB		**27.6**	**28.7**	**33.2**	**47.7**	**57.7**	**64.3**	**67.6**	**64.7**	**56.7**	**47.9**	**36.7**	**31.0**	(26)
(27)	**Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures**	0.4%	WB		**34.5**	**34.2**	**41.5**	**57.1**	**66.2**	**71.2**	**75.3**	**73.0**	**65.6**	**55.9**	**47.7**	**36.4**	(27)
(28)			MCDB		**35.7**	**35.4**	**47.5**	**72.4**	**79.2**	**85.3**	**84.0**	**88.1**	**77.2**	**62.3**	**50.3**	**37.1**	(28)
(29)		2%	WB		**32.6**	**33.1**	**37.2**	**54.2**	**63.3**	**69.1**	**71.7**	**70.3**	**61.8**	**53.0**	**43.8**	**34.2**	(29)
(30)			MCDB		**33.7**	**34.4**	**40.8**	**65.7**	**76.6**	**81.9**	**83.7**	**83.9**	**71.0**	**59.6**	**45.7**	**35.3**	(30)
(31)		5%	WB		**30.3**	**31.6**	**35.2**	**50.9**	**61.1**	**67.3**	**70.1**	**67.9**	**59.5**	**50.3**	**40.1**	**32.6**	(31)
(32)			MCDB		**31.5**	**33.2**	**37.8**	**61.7**	**74.5**	**79.4**	**81.9**	**79.6**	**68.1**	**55.4**	**41.7**	**33.7**	(32)
(33)		10%	WB		**27.6**	**28.7**	**33.6**	**48.0**	**59.0**	**65.5**	**68.5**	**65.8**	**57.5**	**48.2**	**36.8**	**31.0**	(33)
(34)			MCDB		**28.8**	**30.1**	**35.8**	**57.7**	**70.1**	**76.2**	**79.3**	**75.0**	**65.6**	**52.5**	**38.2**	**32.3**	(34)
(35)	**Mean Daily Temperature Range**		MDBR		**10.7**	**12.0**	**13.0**	**15.8**	**18.4**	**17.4**	**17.6**	**16.9**	**15.0**	**10.7**	**7.7**	**9.2**	(35)
(36)		5% DB	MCDBR		**8.6**	**8.5**	**13.1**	**23.4**	**24.5**	**22.7**	**23.3**	**23.8**	**21.8**	**19.1**	**8.4**	**6.2**	(36)
(37)			MCWBR		**8.3**	**7.9**	**9.6**	**11.8**	**11.3**	**10.0**	**8.9**	**9.9**	**10.2**	**11.4**	**7.5**	**5.9**	(37)
(38)		5% WB	MCDBR		**9.2**	**8.0**	**11.5**	**22.4**	**23.6**	**21.1**	**20.1**	**21.9**	**19.3**	**16.8**	**8.2**	**6.2**	(38)
(39)			MCWBR		**9.0**	**7.7**	**9.1**	**12.2**	**11.7**	**10.2**	**9.3**	**10.5**	**10.7**	**11.0**	**7.7**	**6.1**	(39)
(40)	**Clear-Sky Solar Irradiance**		taub		**0.297**	**0.321**	**0.373**	**0.414**	**0.398**	**0.392**	**0.412**	**0.436**	**0.381**	**0.356**	**0.319**	**0.294**	(40)
(41)			taud		**2.030**	**2.025**	**1.957**	**2.087**	**2.283**	**2.350**	**2.298**	**2.205**	**2.335**	**2.284**	**2.164**	**2.016**	(41)
(42)			Ebn at Noon		**180**	**226**	**242**	**253**	**267**	**271**	**261**	**245**	**244**	**216**	**174**	**147**	(42)
(43)			Edh at Noon		**21**	**31**	**43**	**44**	**39**	**36**	**38**	**39**	**30**	**24**	**19**	**17**	(43)
(44)	**All-Sky Solar Radiation**		RadAvg		**227**	**541**	**984**	**1394**	**1726**	**1873**	**1838**	**1427**	**886**	**404**	**169**	**125**	(44)
(45)			RadStd		**31**	**62**	**106**	**122**	**154**	**170**	**165**	**125**	**106**	**80**	**32**	**32**	(45)

Historical Trends

		DBAvg	Heating 99% DB	Heating 99% DP	Cooling 1% DB	Cooling 1% WB	Cooling 1% DP	Degree-Days HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	(46)
(47)	Regional (7 neighbors)	**+1.03**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**+241**	**N/A**	(47)

Nomenclature: See separate page